Math 4610: Fundamentals of Numerical Analysis

Practice Exam — Solutions

1. **Maximize** $\max_{0\le x\le 1}|f(x)|$ **for** $f(x)=e^x+3x$**.**

   $f'(x)=e^x+3>0$ on $[0,1]$, so f is increasing. Since $f(x)>0$ on $[0,1]$, $\max|f(x)|=\max f(x)=f(1)=e+3$.

2. **Taylor polynomial and error bound for** $f(x)=xe^x+x$ **about** $x_0=0$**.**

   (a) $f(0)=0$. $f'(x)=e^x(1+x)+1\Rightarrow f'(0)=2$. $\quad f''(x)=e^x(2+x)\Rightarrow f''(0)=2$.

   $$P_2(x)=f(0)+f'(0)x+\frac{f''(0)}{2}x^2=2x+x^2.$$

   (b) Remainder: $R_2(x)=\dfrac{f^{(3)}(\xi)}{3!}x^3$, for some $\xi\in(0,x)$. Here $f^{(3)}(x)=e^x(x+3)$, so on $[0,1]$,

   $$|f^{(3)}(x)|\le e\cdot 4=4e \quad\Rightarrow\quad |P_2(x)-f(x)|\le\frac{4e}{6}|x|^3\le\frac{2e}{3}\ \text{(for } x\in[0,1]).$$

3. **IEEE 754 double (64-bit) value for**

   $$\underbrace{0}_{\text{sign}}\ \underbrace{00000000011}_{\text{exponent}=3}\ \underbrace{1100000\ldots0}_{\text{fraction}}$$

   Exponent $E=3$, bias $1023\Rightarrow$ actual exponent $e=3-1023=-1020$. Significand $=1+2^{-1}+2^{-2}=1.75=\frac{7}{4}$.

   $$x=(+)\,(1.75)\cdot 2^{-1020}=\ 7\cdot 2^{-1022}\ \approx\ 1.5576\times 10^{-307}.$$

4. **Three-digit rounding arithmetic for** $-10\pi+3$**.**

   Use $\pi\approx 3.14$ (3-digit rounding). Then $10\pi\approx 31.4$, so $-10\pi\approx -31.4$. Now $-31.4+3.00=-28.4$ (already 3 digits).

   $$\boxed{-28.4}$$

5. **Stable formulas for quadratic** $1.002x^2-1000.0x+0.25=0$**.**

   Let $a=1.002$, $b=-1000.0$, $c=0.25$. Then $\sqrt{b^2-4ac}\approx 1000$.

- For x_1 (the "+" root),
$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$
is stable (large numerator); the alternative $\frac{-2c}{b+\sqrt{\cdot}}$ suffers catastrophic cancellation since $b + \sqrt{\cdot} \approx 0$.
- For x_2 (the "−" root),
$$x_2 = \frac{-2c}{b - \sqrt{b^2 - 4ac}}$$
is stable (denominator ≈ -2000), whereas $\frac{-b-\sqrt{\cdot}}{2a}$ has severe cancellation ($1000 - 1000$).

6. **Bisection: $f(x) = \sqrt{x} - \cos x$ on $[0, 1]$, find P_2.**
$f(0) = -1 < 0$, $f(1) = 1 - \cos 1 \approx 0.4597 > 0$.
$$\text{Iter 1: } m_1 = \tfrac{0+1}{2} = 0.5, \quad f(0.5) \approx 0.7071 - 0.8776 < 0 \Rightarrow [0.5, 1].$$
$$\text{Iter 2: } m_2 = \tfrac{0.5+1}{2} = 0.75, \quad f(0.75) \approx 0.8660 - 0.7317 > 0 \Rightarrow P_2 = 0.75.$$

7. **Fixed-point iteration for $g(x) = \pi + 0.5 \sin x$ on $[0, \pi]$, $|p - p_n| \le 10^{-2}$.**
$g'(x) = 0.5 \cos x$, so $L = \max |g'| \le 0.5$ (contraction). Using the a priori bound
$$|p - p_n| \le \frac{L^n}{1 - L} |x_1 - x_0| \le \frac{(0.5)^n}{0.5} \pi = \pi \cdot 2^{1-n} \le 10^{-2}.$$
We need $2^{1-n} \le 10^{-2}/\pi \Rightarrow n \ge 10$. Hence $\boxed{n = 10 \text{ iterations}}$.

8. **Newton's method for $f(x) = -x^3 - \cos x$, $P_0 = -1$, find P_2.**
$f'(x) = -3x^2 + \sin x$.
$$P_1 = P_0 - \frac{f(P_0)}{f'(P_0)} = -1 - \frac{1 - \cos 1}{-3 + \sin(-1)} \approx -1 - \frac{0.4597}{-3.8415} \approx -0.88033.$$
$$P_2 = P_1 - \frac{f(P_1)}{f'(P_1)} \approx -0.88033 - \frac{0.04535}{-3.0959} \approx \boxed{-0.86568}.$$

9. (a) **Degree-2 Lagrange interpolation for $f(0.22)$.**
Use nodes closest to 0.22: $x_0 = 0.1$ ($f = 0.665$), $x_1 = 0.2$ ($f = 0.8$), $x_2 = 0.3$ ($f = 1.8$).
$$P_2(x) = \sum_{k=0}^{2} f(x_k)\, \ell_k(x), \quad \ell_k(x) = \prod_{\substack{j=0 \\ j \ne k}}^{2} \frac{x - x_j}{x_k - x_j}.$$
Evaluating at $x = 0.22$ gives
$$P_2(0.22) \approx \boxed{0.9308}.$$

(b) If $|f^{(3)}(x)| < 5$ on the interval,

$$|f(0.22) - P_2(0.22)| \le \frac{\max|f^{(3)}|}{3!}\,|(0.22-0.1)(0.22-0.2)(0.22-0.3)| \le \frac{5}{6}\cdot 1.92\times 10^{-4} \approx \boxed{1.6 \times 10^{-4}}.$$

10. (a) **Two-point difference (uniform $h = 0.1$) to fill the table.**
Use the forward-difference relation at $x = 0.5$ for the given $f'(0.6)$:

$$f'(0.6) \approx \frac{f(0.6) - f(0.5)}{0.1} \;\Rightarrow\; f(0.6) \approx f(0.5) + 0.1\, f'(0.6) = 1.9 + 0.1 \cdot 2.2 = \boxed{2.12}.$$

(b) **Three-point (most accurate) for $f'(0.6)$; and central for $f''(0.6)$.**
With $x = 0.5, 0.6, 0.7$ and $h = 0.1$:

$$f'(0.6) \approx \frac{f(0.7) - f(0.5)}{2h} = \frac{1.5 - 1.9}{0.2} = \boxed{-2.0}.$$

$$f''(0.6) \approx \frac{f(0.5) - 2f(0.6) + f(0.7)}{h^2} = \frac{1.9 - 2(2.3) + 1.5}{0.01} = \boxed{-120}.$$

11. **Degree of accuracy of $\displaystyle\int_0^a f(x)\,dx \approx \frac{a}{2}f(0) + \frac{a}{2}f(a)$.**

This is the trapezoidal rule on $[0, a]$, exact for constants and linear functions:

$$\int_0^a 1\,dx = a = \frac{a}{2}(1) + \frac{a}{2}(1), \qquad \int_0^a x\,dx = \frac{a^2}{2} = \frac{a}{2}(0) + \frac{a}{2}(a).$$

Not exact for x^2: $\int_0^a x^2 dx = \frac{a^3}{3} \neq \frac{a}{2}\,a^2 = \frac{a^3}{2}$.
Degree of accuracy $= 1$.

12. **Simpson's rule for $\displaystyle\int_{-0.5}^{1.5} x\ln(x+2)\,dx$.**

Use $n = 2$ subintervals, $h = (1.5 - (-0.5))/2 = 1$, nodes $x_0 = -0.5$, $x_1 = 0.5$, $x_2 = 1.5$.

$$f(x) = x\ln(x+2): \quad f(-0.5) = -\tfrac{1}{2}\ln 1.5, \quad f(0.5) = \tfrac{1}{2}\ln 2.5, \quad f(1.5) = \tfrac{3}{2}\ln 3.5.$$

$$S = \frac{h}{3}\big[f(x_0) + 4f(x_1) + f(x_2)\big] \approx \boxed{1.169664454}.$$

13. **Error bound for Simpson's rule above.**

Simpson's error:

$$|E| \le \frac{(b-a)}{180}\, h^4 \max_{[a,b]} \left|f^{(4)}(x)\right|, \quad a=-0.5,\ b=1.5,\ n=2,\ h=1.$$

For $f(x) = x\ln(x+2)$, with $u = x+2$,

$$f''(x) = \frac{1}{u} + \frac{2}{u^2}, \quad f^{(3)}(x) = -\frac{1}{u^2} - \frac{4}{u^3}, \quad f^{(4)}(x) = \frac{2}{u^3} + \frac{12}{u^4}.$$

On $[-0.5, 1.5]$, $u \in [1.5, 3.5]$, so the maximum occurs at $u = 1.5$:

$$M = \max|f^{(4)}| = \frac{2}{1.5^3} + \frac{12}{1.5^4} = \frac{80}{27} \approx 2.963.$$

Hence

$$|E| \le \frac{(1.5-(-0.5))}{180} \cdot 1^4 \cdot M = \frac{2}{180} M \approx \boxed{3.29 \times 10^{-2}}.$$